
Siyu Chen

Ph.D. Candidate in Mechanical Engineering

Villanova University

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RESEARCH PROFILE

Biomechanics and biomedical engineering researcher studying interactions among medical devices, soft biological materials, fluids, and particles. My work combines experimental mechanics, computational modeling, medical imaging, and data-driven analysis, with a central focus on deep brain stimulation and probe–tissue mechanics.

Deep brain stimulation: probe insertion, tissue deformation, interface mechanics, and electrode deviation.

Fluid–structure interaction: oscillatory flow and deformation in compliant tubes, including internal obstructions.

Particle transport: flow-driven assembly and acoustically assisted nanoparticle deposition.

EDUCATION

Ph.D. in Mechanical Engineering | 2022–present

Villanova University. Ph.D. candidate · Advisor: Prof. Qianhong Wu

M.Sc. in Mechanical Engineering | 2019–2021

The Hong Kong Polytechnic University. Concentration: Experimental Fluid Mechanics

B.Sc. in Theoretical and Applied Mechanics | 2014–2018

Southern University of Science and Technology

RESEARCH EXPERIENCE AND COLLABORATION

Doctoral research | 2022–present

Villanova University · Mechanical Engineering. Experimental and computational study of soft material deformation and fluid–structure interaction under Prof. Qianhong Wu, with a focus on DBS probe–tissue mechanics.

Deep brain stimulation biomechanics | **Research collaboration**

Thomas Jefferson University · University of Delaware. Collaborative work with Chengyuan Wu, Feroze Mohamed, and Curtis Johnson on probe–tissue interaction, electrode deviation, and gel characterization.

AWARDS AND HONORS

2023 | Scholarship — Philadelphia Society of Tribologists and Lubrication Engineers (STLE)

2020 | Departmental MSc Dissertation Scholarship — The Hong Kong Polytechnic University

RELEVANT SKILLS

Programming: Python for data analysis and visualization.

Numerical simulation: ANSYS Fluent; COMSOL Multiphysics.

JOURNAL PUBLICATIONS

Bold name identifies Siyu Chen. * Equal contribution.

- J1. Zhou, B.; Ziechmann, R.; **Chen, S.**; Mann, K.; Kramer, M. K.; Kroen, M. B.; Alizadeh, M.; Mohamed, F. B.; Johnson, C. L.; Wu, Q.; Wu, C. (2026). Re-evaluating sources of electrode deviation in deep brain stimulation: surgical technique versus patient-specific brain mechanics. *Journal of Neurosurgery*, 1–9. Published online September 18, 2026. <https://doi.org/10.3171/2026.3.JNS252633>
- J2. **Chen, S.**; Wu, C.; Anders, E.; Nathan, R.; Bailey, O. M.; Wang, Q.; Johnson, C. L.; Wu, Q. (2026). Interface geometry governs probe insertion mechanics and electrode deviation in layered brain-mimicking phantoms for deep brain stimulation. *Biomechanics and Modeling in Mechanobiology*, 25(5), 106. Published online September 16, 2026. <https://doi.org/10.1007/s10237-026-02122-1>
- J3. **Chen, S.**; Nathan, R.; Sidnawi, B.; Anders, E.; Bailey, O. M.; Johnson, C. L.; Wu, C.; Wu, Q. (2026). Biomechanical study of probe–tissue interactions during deep brain stimulation: Force measurement and channel deformation in agar gel phantoms. *Journal of the Mechanical Behavior of Biomedical Materials*, 183, 107582. Published online August 8, 2026. <https://doi.org/10.1016/j.jmbbm.2026.107582>
- J4. Wang, Q.*; **Chen, S.***; Muller, J.; Marchetta, S.; Anders, E.; Hu, H.; Nathan, R.; Wu, C.; Ural, A.; Wu, Q. (2026). Experimentally validated CEL finite element modeling of probe insertion and cavity formation in agar gel for DBS applications. *Computers in Biology and Medicine*, 210, 111732. Published online May 11, 2026. <https://doi.org/10.1016/j.compbiomed.2026.111732>
- J5. Wu, J.*; Zhao, L.*; Dubrunfaut, E. H.; Lallo, V. A.; **Chen, S.**; Wu, Q.; Li, B.; Bracaglia, L. G. (2025). Acoustic Field Enabled Polymeric Nanoparticle Deposition onto Vessel Walls for Enhanced Drug Delivery. *Nano Letters*, 25(23), 9270–9276. Published online May 23, 2025. <https://doi.org/10.1021/acs.nanolett.5c01438>
- J6. Sidnawi, B.; **Chen, S.**; Wang, Q.; Nathan, R.; Wu, Q. (2025). A theoretical and experimental study of an oscillatory flow through a compliant tube. *Physics of Fluids*, 37(4), 041904. Published online April 8, 2025. <https://doi.org/10.1063/5.0264380>

POSTERS

- P1. Zhou, B.; Ziechmann, R.; **Chen, S.**; Mann, K.; Kramer, M. K.; Kroen, M.; Alizadeh, M.; Mohamed, F. B.; Johnson, C. L.; Wu, Q.; Wu, C. (2026). Re-evaluating Sources of Electrode Deviation in Deep Brain Stimulation: Surgical Technique vs. Patient-Specific Brain Mechanics. *NANS 2026 Annual Meeting*. January 22–25, 2026 · Las Vegas, Nevada.
- P2. Sidnawi, B.; **Chen, S.**; Zhao, L.; Li, B.; Wu, Q. (2023). A Statistical Evaluation of the Various Parameters Governing the Assembly of Solution-Based Nanoparticles on Polymer Substrates. *Philadelphia STLE Technical Session*. February 2023.

CONFERENCE PRESENTATIONS

- T1. **Chen, S.**; Ahmed, S.; Sidnawi, B.; Ding, Y.; Li, B.; Wu, Q. (2025). A computational and experimental study of Flow-Driven Assembly of Nanoparticles on Polymer Substrates. *78th Annual Meeting of the Division of Fluid Dynamics*.
- T2. **Chen, S.**; Nathan, R.; Sidnawi, B.; Wu, Q. (2025). On the study of fluid flow through a compliant tube with internal obstructions: Coupled experimental and numerical investigation. *78th Annual Meeting of the Division of Fluid Dynamics*.
- T3. **Chen, S.**; Nathan, R.; Sidnawi, B.; Wang, Q.; Wu, C.; Ural, A.; Mohamed, F.; Johnson, C.; Wu, Q. (2024). Experimental Investigation of Biomechanical Properties Influencing Electrode Deviation in Deep Brain Stimulation. *77th Annual Meeting of the Division of Fluid Dynamics*.
- T4. Wu, Q.; **Chen, S.**; Sidnawi, B.; Nathan, R.; Wang, Q. (2024). On the study of fluid flow through a compliant tube with internal obstructions. *77th Annual Meeting of the Division of Fluid Dynamics*.
- T5. **Chen, S.**; Nathan, R.; Sidnawi, B.; Wang, Q.; Wu, C.; Ural, A.; Mohamed, F.; Johnson, C.; Wu, Q. (2023). A Comprehensive Experimental Study on the Biomechanics of Deep Brain Stimulation. *76th Annual Meeting of the Division of Fluid Dynamics*.
- T6. Sidnawi, B.; **Chen, S.**; Wang, Q.; Nathan, R.; Wu, Q. (2023). A Theoretical and Experimental Study of an Oscillatory Flow Through a Compliant Tube. *76th Annual Meeting of the Division of Fluid Dynamics*.
- T7. Sidnawi, B.; Zhao, L.; **Chen, S.**; Li, B.; Wu, Q. (2023). A Statistical Evaluation of the Various Parameters Governing the Assembly of Solution-Based Nanoparticles on Polymer Substrates. *76th Annual Meeting of the Division of Fluid Dynamics*.
- T8. **Chen, S.**; Muller, J.; Sang, W.; Nathan, R.; Wu, C.; Ural, A.; Wu, Q. (2022). A comprehensive computational and experimental study of deep brain stimulation. *75th Annual Meeting of the Division of Fluid Dynamics*.
- T9. Sidnawi, B.; **Chen, S.**; Wang, Q.; Nathan, R.; Wu, Q. (2022). A Theoretical and Experimental Study of an Oscillatory Flow Through a Compliant Tube. *75th Annual Meeting of the Division of Fluid Dynamics*.